

Protocols and Servers

Telnet

A Telnet server uses the Telnet protocol to listen for incoming connections on port 23. (Please note that the Telnet port is not open on the target VM.) Let's consider the example shown below. A user is connecting to the `telnetd`, a Telnet server. The steps are as follows:

1. First, he is asked to provide his login name (username). We can see the user entering `frank`.
2. Then, he is asked for the password, `D2xc9CgD`. The password is not shown on the screen; however, we display it below for demonstration purposes.
3. Once the system checks his login credentials, he is greeted with a welcome message.
4. And the remote server grants him a command prompt, `frank@bento:~$`. The `$` indicates that this is not a root terminal.

```
Pentester Terminal

pentester@TryHackMe$ telnet 10.129.166.174
Trying 10.129.166.174...
Connected to 10.129.166.174.
Escape character is '^['.
Ubuntu 20.04.3 LTS
bento login: frank
Password: D2xc9CgD
Welcome to Ubuntu 20.04.3 LTS (GNU/Linux 5.4.0-84-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

System information as of Fri 01 Oct 2021 12:24:56 PM UTC

System load:  0.05          Processes:           243
Usage of /:   45.7% of 6.53GB Users logged in:          1
Memory usage: 15%          IPv4 address for ens33: 10.129.166.174
Swap usage:   0%

 * Super-optimized for small spaces - read how we shrank the memory
   footprint of MicroK8s to make it the smallest full K8s around.

https://ubuntu.com/blog/microk8s-memory-optimisation

0 updates can be applied immediately.

*** System restart required ***
Last login: Fri Oct  1 12:17:25 UTC 2021 from meiyu on pts/3
You have mail.
frank@bento:~$
```

HTTP

web browser connects to the webserver and uses HTTP to request HTML pages and images among other files and submit forms and upload various files

HTTP



HTTP sends and receives data as cleartext (not encrypted); therefore, you can use a simple tool, such as Telnet (or Netcat), to communicate with a web server and act as a “web browser”. The key difference is that you need to input the HTTP-related commands instead of the web browser doing that for you.

Telnet is a simple protocol; furthermore, it uses cleartext for communication. We will use `telnet` instead of a web browser to request a file from the webserver. The steps will be as follows:

1. First, we connect to port 80 using `telnet MACHINE_IP 80`.
2. Next, we need to type `GET /index.html HTTP/1.1` to retrieve the page `index.html` or `GET / HTTP/1.1` to retrieve the default page.
3. Finally, you need to provide some value for the host like `host: telnet` and press the Enter/Return key **twice**.

```
pentester@TryHackMe$ telnet MACHINE_IP 80
Trying MACHINE_IP...
Connected to MACHINE_IP.
Escape character is '^]'.
GET /index.html HTTP/1.1
host: telnet

HTTP/1.1 200 OK
Server: nginx/1.18.0 (Ubuntu)
Date: Wed, 15 Sep 2021 08:56:20 GMT
Content-Type: text/html
Content-Length: 234
Last-Modified: Wed, 15 Sep 2021 08:53:59 GMT
Connection: keep-alive
ETag: "6141b4a7-ea"
Accept-Ranges: bytes

<!DOCTYPE html>
<html lang="en">
<head>
  <title>Welcome to my Web Server</title>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width,initial-scale=1" />
</head>
<body>
  <h1>Coming Soon</h1>
</body>
</html>
```

Three popular choices for HTTP servers are:

- [Apache\(opens in new tab\)](#)
- [Internet Information Services \(IIS\)\(opens in new tab\)](#)
- [nginx](#)

There are many web browsers available. At the time of writing, the most popular web browsers are:

- Chrome by Google
- Edge by Microsoft
- Firefox by Mozilla
- Safari by Apple.

Practical example:

```
root@ip-10-130-103-46:~# telnet 10.130.134.60 80
Trying 10.130.134.60...
Connected to 10.130.134.60.
Escape character is '^]'.
GET /flag.thm HTTP/1.1
host: telnet

HTTP/1.1 200 OK
Server: nginx/1.18.0 (Ubuntu)
Date: Thu, 30 Apr 2026 15:59:49 GMT
Content-Type: application/octet-stream
Content-Length: 39
Last-Modified: Wed, 15 Sep 2021 09:19:23 GMT
Connection: keep-alive
ETag: "6141ba9b-27"
Accept-Ranges: bytes

THM{e3eb0a1df437f3f97a64aca5952c8ea0}
```

FTP

File Transfer Protocol (FTP) was developed to make the transfer of files between different computers with different systems efficient.

FTP also sends and receives data as cleartext; therefore, we can use Telnet (or Netcat) to communicate with an FTP server and act as an FTP client. In the example below, we carried out the following steps:

1. We connected to an FTP server using a Telnet client. Since FTP servers listen on port 21 by default, we had to specify to our Telnet client to attempt connection to port 21 instead of the default Telnet port.
2. We needed to provide the username with the command `USER frank`.
3. Then, we provided the password with the command `PASS D2xc9CgD`.
4. Because we supplied the correct username and password, we got logged in.

A command like `STAT` can provide some added information. The `SYST` command shows the System Type of the target (UNIX in this case). `PASV` switches the mode to passive. It is worth noting that there are two modes for FTP:

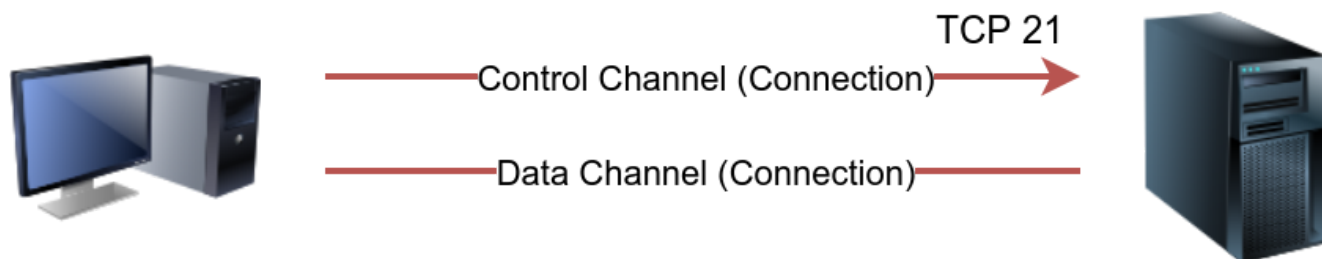
- Active: In the active mode, the data is sent over a separate channel originating from the FTP server's port 20.
- Passive: In the passive mode, the data is sent over a separate channel originating from an FTP client's port above port number 1023.

TYPE A switches the file transfer mode to ASCII, while TYPE I switches the file transfer mode to binary.

example:

```
pentester@TryHackMe$ telnet 10.130.134.60 21
Trying 10.130.134.60...
Connected to 10.130.134.60.
Escape character is '^]'.
220 (vsFTPd 3.0.3)
USER frank
331 Please specify the password.
PASS D2xc9CgD
230 Login successful.
SYST
215 UNIX Type: L8
PASV
227 Entering Passive Mode (10,10,0,148,78,223).
TYPE A
200 Switching to ASCII mode.
STAT
211-FTP server status:
    Connected to ::ffff:10.10.0.1
    Logged in as frank
    TYPE: ASCII
    No session bandwidth limit
    Session timeout in seconds is 300
    Control connection is plain text
    Data connections will be plain text
    At session startup, client count was 1
    vsFTPd 3.0.3 - secure, fast, stable
211 End of status
QUIT
221 Goodbye.
Connection closed by foreign host.
```

FTP



small number of commands to retrieve the file. After logging in successfully, we get the FTP prompt, `ftp>`, to execute various FTP commands. We used `ls` to list the files and learn the file name; then, we switched to `ascii` since it is a text file (not binary). Finally, `get` `FILENAME` made the client and server establish another channel for file transfer.

```
pentester@TryHackMe$ ftp 10.130.134.60
Connected to 10.130.134.60.
220 (vsFTPd 3.0.3)
Name: frank
331 Please specify the password.
Password: D2xc9CgD
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp> ls
227 Entering Passive Mode (10,20,30,148,201,180).
150 Here comes the directory listing.
-rw-rw-r--  1 1001    1001    4006 Sep 15 10:27 README.txt
226 Directory send OK.
ftp> ascii
200 Switching to ASCII mode.
ftp> get README.txt
local: README.txt remote: README.txt
227 Entering Passive Mode (10,10,0,148,125,55).
150 Opening BINARY mode data connection for README.txt (4006 bytes).
WARNING! 9 bare linefeeds received in ASCII mode
File may not have transferred correctly.
226 Transfer complete.
4006 bytes received in 0.000269 secs (14892.19 Kbytes/sec)
ftp> exit
221 Goodbye.
```

Examples of FTP server software include:

- [vsftpd\(opens in new tab\)](#)
- [ProFTPD\(opens in new tab\)](#)
- [uFTP](#)

FTP client commonly found on Linux systems, you can use an FTP client with GUI such as [FileZilla\(opens in new tab\)](#). Some web browsers also support FTP protocol.

to connect to ftp can use `ftp` command with the ip of the machine/server/website then with login credentials can access ftp then use linux commands

```

root@ip-10-130-103-46:~# ftp 10.130.134.60
Connected to 10.130.134.60.
220 (vsFTPd 3.0.5)
Name (10.130.134.60:root): frank
331 Please specify the password.
Password:
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp> ls
200 PORT command successful. Consider using PASV.
150 Here comes the directory listing.
drwx-----  10 1001    1001    4096 Sep 15  2021 Maildir
-rw-rw-r--   1 1001    1001    4006 Sep 15  2021 README.txt
-rw-rw-r--   1 1001    1001      39 Sep 15  2021 ftp_flag.thm
226 Directory send OK.
ftp> get README.txt
local: README.txt remote: README.txt
200 PORT command successful. Consider using PASV.
150 Opening BINARY mode data connection for README.txt (4006 bytes).
226 Transfer complete.
4006 bytes received in 0.00 secs (3.5705 MB/s)
ftp> ascii
200 Switching to ASCII mode.
ftp> get README.txt
local: README.txt remote: README.txt
200 PORT command successful. Consider using PASV.
150 Opening BINARY mode data connection for README.txt (4006 bytes).
WARNING! 9 bare linefeeds received in ASCII mode
File may not have transferred correctly.
226 Transfer complete.
4006 bytes received in 0.00 secs (19.1021 MB/s)
ftp> get ftp_flag.thm
local: ftp_flag.thm remote: ftp_flag.thm
200 PORT command successful. Consider using PASV.
150 Opening BINARY mode data connection for ftp_flag.thm (39 bytes).
WARNING! 2 bare linefeeds received in ASCII mode
File may not have transferred correctly.
226 Transfer complete.
39 bytes received in 0.00 secs (52.9707 kB/s)

```

I retrieved the file and it ended up in the root directory so opened another terminal to read the

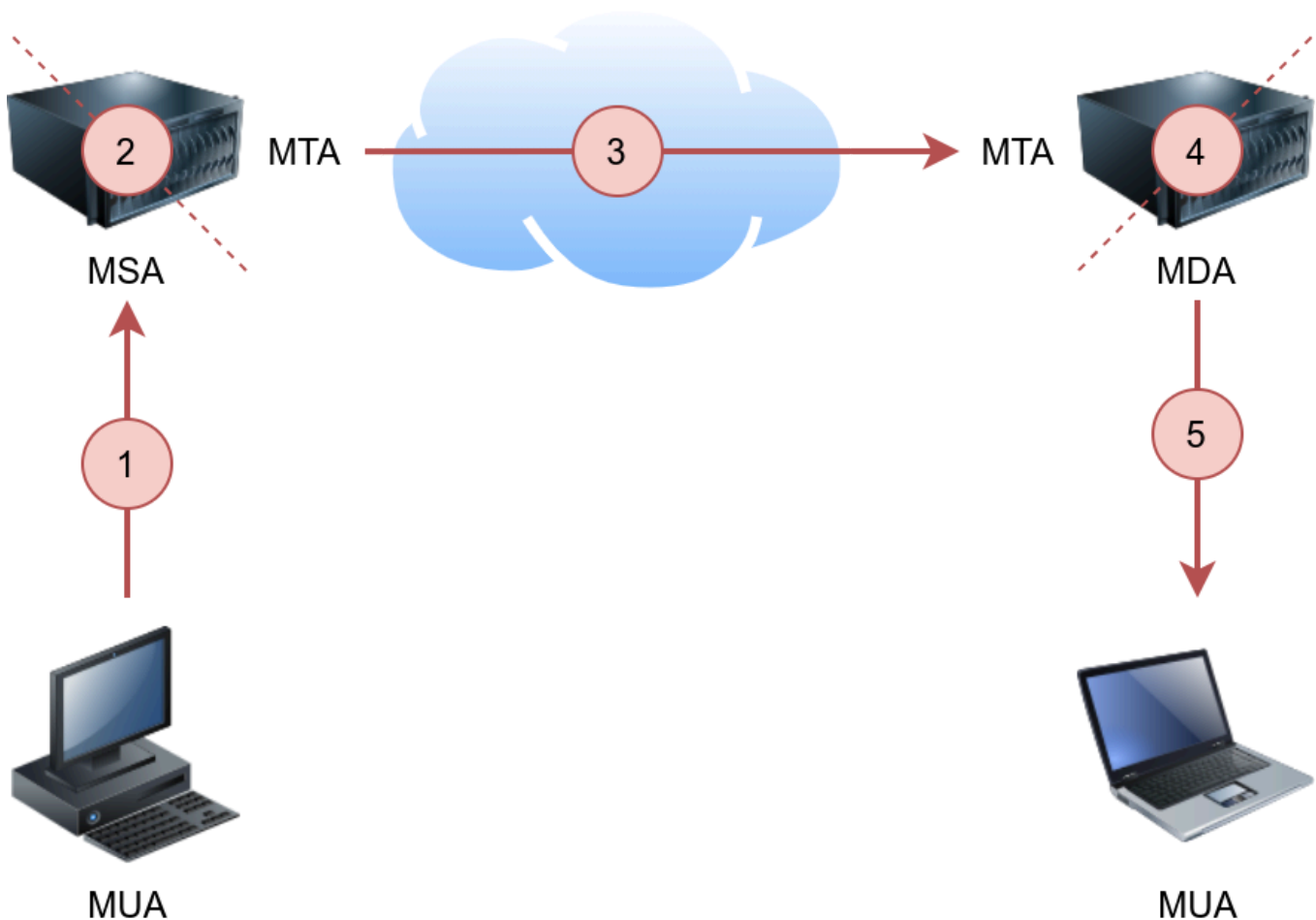
file and get the flag

```
root@ip-10-130-103-46: ~
File Edit View Search Terminal Help
root@ip-10-130-103-46:~# ls
burp.json  Downloads  Pictures  Rooms  thinclient_drives
CTFBuilder ftp_flag.thm Postman  Scripts Tools
Desktop    Instructions README.txt snap
root@ip-10-130-103-46:~# cat ftp_flag.thm
THM{364db6ad0e3ddfe7bf0b1870fb06fbdf}
root@ip-10-130-103-46:~#
```

SMTP

Email delivery over the Internet requires the following components:

1. Mail Submission Agent (MSA)
2. Mail Transfer Agent (MTA)
3. Mail Delivery Agent (MDA)
4. Mail User Agent (MUA)



1. A Mail User Agent (MUA), or simply an email client, has an email message to be sent. The MUA connects to a Mail Submission Agent (MSA) to send its message.
2. The MSA receives the message, checks for any errors before transferring it to the Mail Transfer Agent (MTA) server, commonly hosted on the same server.
3. The MTA will send the email message to the MTA of the recipient. The MTA can also function as a Mail Submission Agent (MSA).
4. A typical setup would have the MTA server also functioning as a Mail Delivery Agent (MDA).
5. The recipient will collect its email from the MDA using their email client.

In the same way, we need to follow a protocol to communicate with an HTTP server, and we need to rely on email protocols to talk with an MTA and an MDA. The protocols are:

1. Simple Mail Transfer Protocol (SMTP)
2. Post Office Protocol version 3 (POP3) or Internet Message Access Protocol (IMAP)

Simple Mail Transfer Protocol (SMTP) is used to communicate with an MTA server. Because SMTP uses cleartext, where all commands are sent without encryption, we can use a basic Telnet client to connect to an SMTP server and act as an email client (MUA) sending a message.

SMTP server listens on port 25 by default

To see basic communication with an SMTP server, we used Telnet to connect to it. Once connected, we issue `helo hostname` and then start typing our email.

```
pentester@TryHackMe$ telnet 10.130.134.60 25
Trying 10.130.134.60...
Connected to 10.130.134.60.
Escape character is '^]'.
220 bento.localdomain ESMTP Postfix (Ubuntu)
helo telnet
250 bento.localdomain
mail from:
250 2.1.0 Ok
rcpt to:
250 2.1.5 Ok
data
354 End data with .
subject: Sending email with Telnet
Hello Frank,
I am just writing to say hi!
.
250 2.0.0 Ok: queued as C3E7F45F06
quit
221 2.0.0 Bye
Connection closed by foreign host.
```

After `helo`, we issue `mail from:`, `rcpt to:` to indicate the sender and the recipient. When we send our email message, we issue the command `data` and type our message. We issue `<CR><LF>.<CR><LF>` (or `Enter . Enter` to put it in simpler terms)

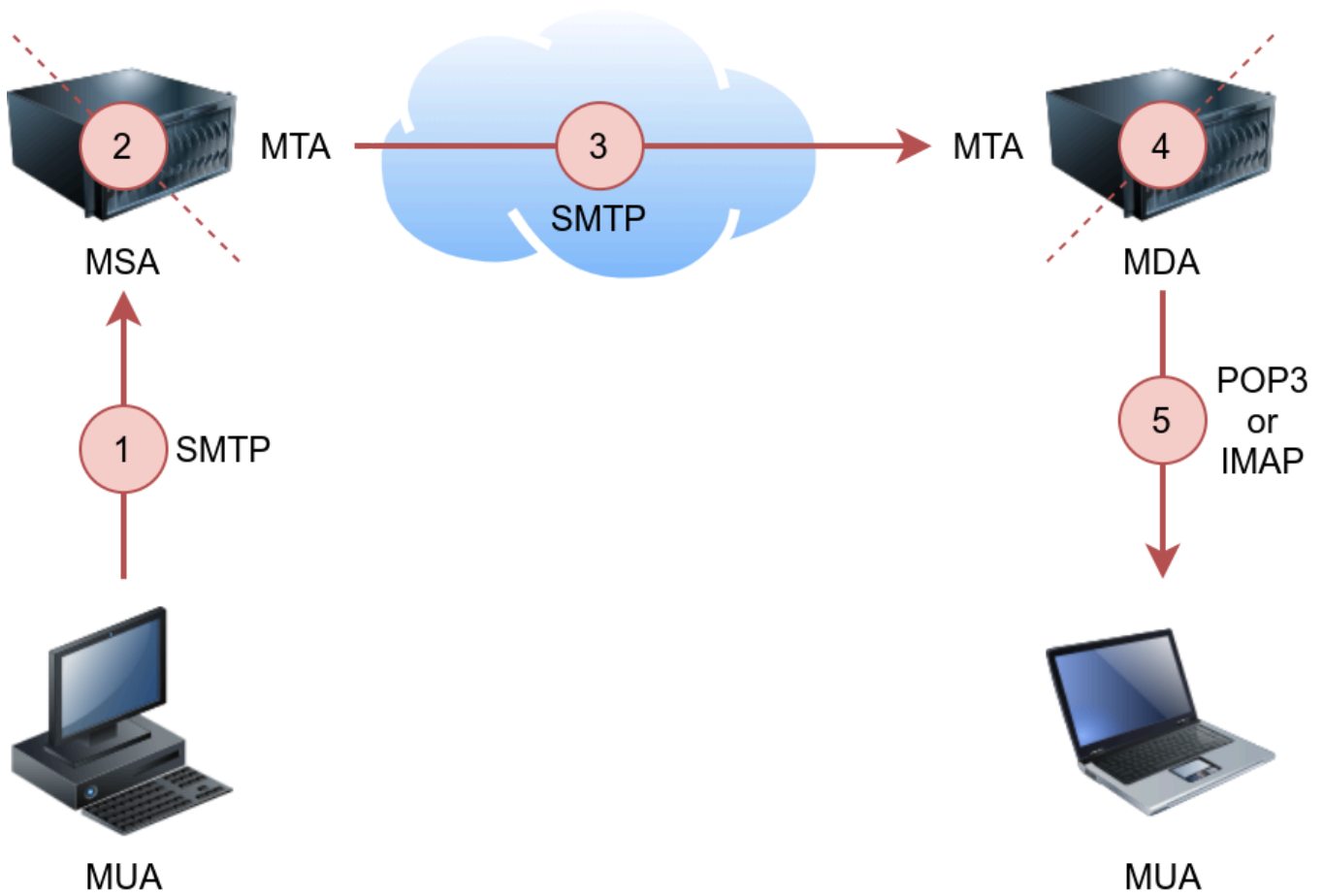
Practical example:

i just used the telnet command with the smtp port to get the flag

```
root@ip-10-130-103-46:~# telnet 10.130.134.60 25
Trying 10.130.134.60...
Connected to 10.130.134.60.
Escape character is '^]'.
220 bento.localdomain ESMTP Postfix THM{5b31ddfc0c11d81eba776e983c35e9b5}
^C
```

POP3

protocol used to download the email messages from a Mail Delivery Agent (MDA) server



POP3 default port 110

process:

use telnet IP 110 (pop3 portnumber)

login with credentials USER:

PASS:

Then can use commands like STAT

LIST

RETR 1

Practical example:

```
root@ip-10-130-103-46:~# telnet 10.130.134.60 110
Trying 10.130.134.60...
Connected to 10.130.134.60.
Escape character is '^]'.
+OK Hello there.
USER frank
+OK Password required.
PASS D2xc9CgD
+OK logged in.
STAT
+OK 0 0
LIST
+OK POP3 clients that break here, they violate STD53.
.
RETR 1
-ERR Invalid message number.
```

STAT and LIST didnt return anything as there is no mail

IMAP

IMAP makes it possible to keep your email synchronized across multiple devices (and mail clients). In other words, if you mark an email message as read when checking your email on your smartphone, the change will be saved on the IMAP server (MDA) and replicated on your laptop when you synchronize your inbox.

Process:

use telnet IP 143 (IMAP} portnumber)

login with credentials:

LOGIN username password

IMAP requires each command to be preceded by a random string to be able to track the reply.

So we added c1 , then c2

can list mail folders using:

LIST "" "[*]"

can also examine inbox:

EXAMINE INBOX

Practical Example:

```
root@ip-10-130-103-46:~# telnet 10.130.134.60 143
Trying 10.130.134.60...
Connected to 10.130.134.60.
Escape character is '^]'.
* OK [CAPABILITY IMAP4rev1 UIDPLUS CHILDREN NAMESPACE THREAD=ORDEREDSUBJECT THREAD=REFERENCES SORT QUOTA IDLE ACL ACL2=UNION STARTTLS ENABLE UTF8=ACCEPT] Courier-IMAP ready. Copyright 1998-2018 Double Precision, Inc. See COPYING for distribution information.
c1 LOGIN frank D2xc9CgD
* OK [ALERT] Filesystem notification initialization error -- contact your mail administrator (check for configuration errors with the FAM/Gamin library)
c1 OK LOGIN ok.
c2 LIST "" "*"
* LIST (\HasNoChildren) "." "INBOX.Trash"
* LIST (\HasNoChildren) "." "INBOX.Drafts"
* LIST (\HasNoChildren) "." "INBOX.Templates"
* LIST (\HasNoChildren) "." "INBOX.Sent"
* LIST (\Unmarked \HasChildren) "." "INBOX"
c2 OK LIST completed
c3 EXAMINE INBOX
* FLAGS (\Draft \Answered \Flagged \Deleted \Seen \Recent)
* OK [PERMANENTFLAGS ()] No permanent flags permitted
* 0 EXISTS
* 0 RECENT
* OK [UIDVALIDITY 777567454] ok
* OK [MYRIGHTS "acdilrsw"] ACL
c3 OK [READ-ONLY] ok
```

Summary

Protocol	TCP Port	Application(s)	Data Security
FTP	21	File Transfer	Cleartext
HTTP	80	Worldwide Web	Cleartext
IMAP	143	Email (MDA)	Cleartext
POP3	110	Email (MDA)	Cleartext
SMTP	25	Email (MTA)	Cleartext
Telnet	23	Remote Access	Cleartext